Discrete Mathematics for Computer Science

B.Tech. (CSE) - First Year, Semester II

Dr. Rishikant Rajdeepak

Assistant Professor
School of Computer Science Engineering & Applications
D Y Patil International University, Akurdi, Pune

AY 2025-26

Course Overview

Course Details

Course Name: Discrete Mathematics for Computer Science

Course Code: CSE1101

Credits: 5

Prerequisites: Intermediate level mathematics

L-T-P-C: 3-1-2-5

Course Objectives

1. To provide a strong foundation in mathematical logic, set theory, and relations for analyzing complex problems.
2. To introduce the concepts of graph theory and algebraic structures as tools for computer science applications.
3. To develop the ability to construct formal mathematical proofs using induction and logical reasoning.
4. To apply discrete structures to represent and solve real-world computational problems.

Course Syllabus — Module I

Module I: Logic and Proof Techniques

Propositions; Logical connectives; Truth tables; Conditional statements, converse, contrapositive, and inverse; Logical equivalence; Tautology; Contradiction; Contingency; Conjunctive normal form; Disjunctive normal form; Translating english sentences; Predicate logic; Universal quantifier; Existential quantifier; Methods of proof; Mathematical induction.

Course Syllabus — Module II

Module II: Sets, Functions, and Relations

Sets; Venn diagrams; Set operations; Subsets; Set identities; Cartesian product of sets; Power set; Cardinality of sets; Basic counting techniques; Principle of inclusion-exclusion; Functions; Injection, surjection, and bijection; Composition of functions; Invertible functions; Relations; Properties of relations; Equivalence relations; Equivalence classes and partitions; Posets; Lattices.

Course Syllabus — Module III

Module III: Graph Theory

Graphs; Types of graphs; Basic terminology in graphs; Handshaking lemma; Paths and cycles; Trees; Complete graphs; Regular graphs; Bipartite and k-partite graphs; Graph representations; Connected graphs; Components of graphs; Euler and Hamiltonian paths; Graph coloring; Chromatic number.

Course Syllabus — Module IV

Module IV: Algebraic structures

Semigroups; Monoids; Groups; Abelian groups; Subgroups; Cyclic groups; Group homomorphism and isomorphism; Rings; Fields;

Laboratory

1. Generate truth tables for simple and compound propositions.
2. Determine logical equivalence among the given propositions using truth tables.
3. Perform set operations on finite sets.
4. Compute the Cartesian product of finite sets and determine its cardinality.
5. Determine whether a given relation is an equivalence relation.
6. Construct a graph from the given edge list and obtain its adjacency matrix.
7. Verify the Handshaking Lemma for a given graph.
8. Construct the Cayley table for a finite set under a given binary operation.

Course Outcomes

At the end of the course the student will be able to:

- Apply rules of propositional logic, predicate logic, and mathematical induction to verify the validity of mathematical arguments and formal proofs.
- Solve problems involving sets, functions, counting techniques, and relations to interpret partitions and equivalence structures.
- Examine graphs to analyze properties such as connectivity, paths, trees, and coloring for structural modeling.
- Implement algebraic structures, including groups, rings, and fields, in suitable mathematical and computational situations.

Textbooks

1. K. H. Rosen, *Discrete Mathematics and its Applications*, 7th ed. McGraw-Hill, 2012.
2. Swapan K. Sarkar, *Textbook of Discrete Mathematics*, 1st ed., 2015.
3. B. Kolman, R. C. Busby, and S. C. Ross, *Discrete Mathematical Structures*, 6th ed., Pearson, 2008.
4. C. L. Liu and D. P. Mohapatra, *Elements of Discrete Mathematics: A Computer Oriented Approach*, 4th ed., Tata McGraw-Hill, 2012.

References

1. NPTEL, "Discrete Mathematics for CS," Prof. Nitin Saxena, IIT Kanpur.
2. MIT OpenCourseWare, "Mathematics for Computer Science," 6.042J.
3. Coursera, "Introduction to Discrete Mathematics for Computer Science Specialization."

Evaluation Scheme

Theory: 60%			
Continuous Assessment	**Mid-term**	**End-term**	**Total**
25	25	50	100

Note: Continuous Assessment within the course will consist only of four quizzes (one per module), contributing 25% to the continuous assessment for theory.

Practical: 40%				
Continuous Assessment			**End-term Practical**	**Total**
Lab Manual	**Lab Assessment**	**Internal Viva-voce**		
10	10	30	50	100

Special Notes

- Please read the 'Student Examination Handbook (2025–2026)' shared by the COE carefully.
- All lectures, tutorials, and study materials will be available in JUNO ERP.
- Maintain a dedicated notebook for this course. Practice regularly and be prepared to present it whenever asked.
- Office hours: Attendance and discussion notes will be kept on record.
  - Div C: Tuesday 09:00 - 10:00
  - Div D: Monday 09:00 - 10:00
- Office: Room 521.

Questions!

Module I: Logic and Proof Techniques

Topics for This Lecture

Propositions • Logical Connectives • Truth Tables

What is Logic?

- Logic is the study of methods and principles used to distinguish correct reasoning from incorrect reasoning.
- In this course, we focus on **propositional logic**.

Basic ideas

- A **proposition** is a declarative sentence that is either true or false, but not both.
- **Propositional logic (propositional calculus)** is the branch of logic that studies propositions and how they combine using logical connectives.

Examples of Propositions

- A proposition must have a definite truth value: **true or false**.

Examples

- $2 + 3 = 5$
- New Delhi is the capital of India.
- 7 is an even number.
- The sun rises in the east.

Each of these statements is either true or false (but not both).

Not Propositions

- Some sentences do **not** have a definite truth value.

Not propositions

- Close the door.
- What time is it?
- $x + 5 = 12$
- Study harder!

Commands, questions, and statements with unknown variables are **not** propositions.

Propositional Variables and Truth Values

- We use letters to represent propositions:

$$p, q, r, s, \ldots$$

  They are called **propositional variables**.
- Each propositional variable has a **truth value**:

$$\text{True (T)} \quad \text{or} \quad \text{False (F)}$$

Examples

- p: 2 is an even number. (T)
- q: 5 is a prime number. (T)
- r: $3 + 4 = 10$. (F)

Compound Propositions

- Propositions can be combined to form **compound propositions**.
- We use logical connectives to build more complex statements.

Example

p : It is raining. q : I am carrying an umbrella.

$p \wedge q$: It is raining and I am carrying an umbrella.

The truth of a compound proposition depends on the truth values of the propositions that form it.

Logical Connectives

Precedence	Symbol	Meaning
1	$\neg p$	Negation: not p
2	$p \wedge q$	Conjunction: p and q
3	$p \vee q$	Disjunction (inclusive or): p or q (or both)
3	$p \oplus q$	Exclusive or (XOR): exactly one of p, q is true
4	$p \rightarrow q$	Implication: if p, then q
5	$p \leftrightarrow q$	Biconditional: p iff q

Note that OR and XOR have same precedence order. Parentheses can always be used to avoid ambiguity.

Truth Tables

- A **truth table** shows the truth value of a compound proposition for every possible combination of truth values of its variables.

Example

- $\neg p$ means "not p".
- It reverses the truth value of p.

p	$\neg p$
T	F
F	T

Truth Table Example with OR

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Observation:

- $p \vee q$ is false only when both p and q are false.

Truth Table Example with AND

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Observation:

- $p \wedge q$ is true only when both p and q are true.

Truth Table Example with Exclusive OR (XOR)

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Observation:

- $p \oplus q$ is true only when either p or q is true (not both).

Implication

- $p \to q$ means "If p, then q".
- It is false only when p is true and q is false.

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional

- $p \leftrightarrow q$ means "p if and only if q".

  p is the hypothesis, and q is the conclusion.
- True when p and q have the same truth value.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example

Let:

p: "It is raining"

q: "I carry an umbrella"

- $\neg p$: It is not raining.
- $p \wedge q$: It is raining and I carry an umbrella.
- $p \vee q$: It is raining or I carry an umbrella (or both).
- $p \oplus q$: Either it is raining or I carry an umbrella, but not both.
- $p \rightarrow q$: If it is raining, then I carry an umbrella.
- $p \leftrightarrow q$: I carry an umbrella if and only if it is raining.

Exercises — Propositions

- Decide whether each of the following is a proposition. If it is, give its truth value.
  1. $2 + 5 = 9$
  2. Close the window.
  3. Every prime number is odd.
  4. $x + 3 = 10$
  5. Paris is the capital of France.

Exercises — Negation

- Write the negation of each statement in simple English.
  1. The class starts at 9:00.
  2. The student submitted the assignment.
  3. The number 7 is even.
  4. All computers in the lab are working.

Exercises — Conjunction and Disjunction

- Express each sentence using symbols and then find its truth value.
  1. It is raining and it is cold.
  2. She is a teacher or she is a doctor.
  3. The bus is late and the road is empty.
  4. He will pass the exam or he will repeat the course.

Exercises — Exclusive OR

- Rewrite each sentence using exclusive or ($\oplus$).
  1. You can choose tea or coffee, but not both.
  2. The room will be locked by Alice or Bob, but not both.
  3. The light is on in exactly one of the two rooms.

Exercises — Implication

- Translate each into symbolic form.
  1. If it rains, the match will be cancelled.
  2. If a number is even, then it is divisible by 2.
  3. If the student studies, then the student passes.

Exercises — Biconditional

- Write each as a biconditional statement.
  1. A figure is a square if and only if it has four equal sides and four right angles.
  2. You may enter the lab if and only if you have your ID card.
  3. A number is even if and only if it is divisible by 2.

Module I: Logic and Proof Techniques

Topics for This Lecture

Conditional statements • Converse • Contrapositive • Inverse

Conditional Statements

- A conditional statement has the form

$$p \to q$$

- Read as: **If p, then q.**
- p is the hypothesis (antecedent)
- q is the conclusion (consequent)

Example

If a number is divisible by 4, then it is even.

Truth of a Conditional

- $p \to q$ is false only when

$$p \text{ is true and } q \text{ is false.}$$

- In every other case, it is true.

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Converse, Inverse, Contrapositive

Given the conditional

$$p \to q$$

- **Converse:** $q \to p$
- **Inverse:** $\neg p \to \neg q$
- **Contrapositive:** $\neg q \to \neg p$

Important

$$p \to q \quad \text{and} \quad \neg q \to \neg p$$

have the same truth value.

Example — Converse, Inverse, Contrapositive

Original statement

If it is raining, then the ground is wet.

- **Converse:** If the ground is wet, then it is raining.
- **Inverse:** If it is not raining, then the ground is not wet.
- **Contrapositive:** If the ground is not wet, then it is not raining.

Logical Equivalence

- Two compound propositions are **logically equivalent** if they have the same truth value in every possible case.

$$p \equiv q \quad \text{means "}p \text{ and } q \text{ are logically equivalent."}$$

Example

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

(This is one of De Morgan's laws.)

Truth Table: Conditional, Converse, Inverse, Contrapositive

p	q	$p \to q$	$q \to p$	$\neg p \to \neg q$	$\neg q \to \neg p$
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

Observation:

- $p \to q \equiv \neg q \to \neg p$.
- $q \to p \equiv \neg p \to \neg q$.

Logical Equivalence Using Truth Tables

$$p \to q \equiv \neg p \vee q$$

p	q	$p \to q$	$\neg p \vee q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Both columns are identical, so the statements are logically equivalent.

Module I: Logic and Proof Techniques

Topics for This Lecture

Logical Equivalence • Tautology • Contradiction • Contigency

Tautology, Contradiction, and Contingency

- A **tautology**: always true.
- A **contradiction**: always false.
- A **contingency**: sometimes true, sometimes false.

$$p \vee \neg p \quad \text{(tautology)}$$

$$p \wedge \neg p \quad \text{(contradiction)}$$

Logical Equivalence

Two propositions are **logically equivalent** if they always have the same truth value.

$$p \equiv q$$

This symbol "$\equiv$" is NOT a logical connective. It means: "these two statements have identical truth tables."

Another way to say the same thing:

$$(p \leftrightarrow q) \text{ is a tautology.}$$

Example

$$p \rightarrow q \equiv \neg p \vee q$$

Logical Equivalences — I

- Identity laws:

$$p \wedge T \equiv p \qquad p \vee F \equiv p$$

- Domination laws:

$$p \vee T \equiv T \qquad p \wedge F \equiv F$$

- Idempotent laws:

$$p \vee p \equiv p \qquad p \wedge p \equiv p$$

- Double negation laws:

$$\neg(\neg p) \equiv p$$

Logical Equivalences — II

- Commutative laws:

$$p \vee q \equiv q \vee p \qquad p \wedge q \equiv q \wedge p$$

- Associative laws:

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

Logical Equivalences — III

- Distributive laws:

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

- De Morgan laws:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Logical Equivalences — IV

- Absorption laws:

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$

- Negation laws:

$$p \vee \neg p \equiv T$$

$$p \wedge \neg p \equiv F$$

Equivalences Involving Conditionals

$$p \to q \equiv \neg p \vee q$$

$$p \to q \equiv \neg q \to \neg p$$

$$\neg(p \to q) \equiv p \wedge \neg q$$

$$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$$

$$(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$$

Equivalences Involving Biconditionals

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Exercises — Logical Equivalence

Do not use truth tables. Use logical equivalence laws only.

- Show that
$$\neg(p \to q) \quad \text{and} \quad p \wedge \neg q$$
are logically equivalent.
- Show that
$$\neg(p \vee (\neg p \wedge q)) \quad \text{and} \quad \neg p \wedge \neg q$$
are logically equivalent by developing a series of logical equivalences.
- Show that
$$(p \wedge q) \to (p \vee q)$$
is a tautology.

Module I: Logic and Proof Techniques

Topics for This Lecture

Conjunctive normal form • Disjunctive normal form

Disjunctive Normal Form (DNF)

A compound proposition is in **disjunctive normal form (DNF)** if it is a:

disjunction of conjunctions of literals

Form:

$$(p \wedge q) \vee (\neg p \wedge r) \vee \cdots$$

Literal: a variable or its negation.

Every compound proposition is logically equivalent to some DNF.

Motivation: Why DNF?

- Truth tables describe where a statement is true.
- DNF gives an algebraic expression for those rows.
- Useful in:
  - logic design
  - Boolean algebra
  - reasoning and proofs

Idea: build a conjunction for each row where the statement is TRUE, then OR them together.

Example: Building DNF from Truth Table

Suppose the statement is true only when

$$p = T,\ q = F \quad \text{or} \quad p = F,\ q = T$$

Row 1: $p = T,\ q = F \Rightarrow p \wedge \neg q$
Row 2: $p = F,\ q = T \Rightarrow \neg p \wedge q$

$$\Rightarrow (p \wedge \neg q) \vee (\neg p \wedge q)$$

This is a DNF expression.

More DNF Examples

Example 1

$$p \vee q$$

is already DNF, since a single disjunction is allowed.

Example 2

$$(p \to q)$$

Rewrite:

$$p \to q \equiv \neg p \vee q$$

This is DNF.

Example 3

$$\neg(p \wedge q)$$

Rewrite:

Exercises — DNF

Convert the following into DNF.

(1) $\neg(p \vee q)$

(2) $(p \wedge q) \vee (\neg p \wedge r)$

(3) $(p \rightarrow q)$

(4) $\neg(p \wedge \neg q)$

Hint: Use De Morgan, distributive, and implication laws.

Conjunctive Normal Form (CNF)

A compound proposition is in **conjunctive normal form (CNF)** if it is a:

conjunction of disjunctions of literals

Form:

$$(p \vee q) \wedge (\neg p \vee r) \wedge \cdots$$

Every compound proposition is logically equivalent to some CNF.

Motivation: Why CNF?

- Used extensively in automated theorem proving.
- SAT solvers work directly with CNF.
- Many logical arguments simplify into CNF.

Idea: build a clause for each row that makes the statement FALSE, then AND them together.

Example: Building CNF

Consider: true except when

$$p = T, \ q = F$$

To force this row to be false, we build:

$$(\neg p \vee q)$$

Thus this single clause already gives CNF.

$$(\neg p \vee q)$$

More CNF Examples

Example 1

$$p \wedge q$$

Already CNF.

Example 2

$$p \rightarrow q$$

Rewrite:

$$p \rightarrow q \equiv \neg p \vee q$$

This is also CNF.

Example 3

$$\neg(p \vee q)$$

Exercises — CNF

Convert to CNF.

(1) $\neg(p \wedge q)$

(2) $(p \vee (q \wedge r))$

(3) $(p \leftrightarrow q)$

(4) $\neg(p \rightarrow q)$

Hint: Push negations inside, then distribute.

Important Observations

- DNF describes when a proposition is TRUE.
- CNF describes when a proposition is FALSE.
- Both are guaranteed to exist for every proposition.
- Neither form is unique — expressions can often be simplified.

Mixed Practice

Rewrite each into BOTH DNF and CNF.

(1) $(p \wedge r) \rightarrow q$

(2) $\neg(p \leftrightarrow q)$

(3) $(p \vee q) \wedge \neg r$

Do not use truth tables unless necessary.

Thank You!